# PDAN8411: POE PART1

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# Dataset suitability for linear regression

The dataset was suitable for linear regression because the variable being predicted, charges, is a continuous numeric value. This matters because linear regression is used to predict numeric outcomes rather than categories (Müller & Guido, 2016). The dataset also included useful input variables such as age, BMI, smoking status, sex, number of children, and region, all of which could reasonably affect medical costs.

Some of the variables were categorical, but they could still be used once they were encoded into numeric form (Müller & Guido, 2016). Before modelling, the dataset still needed to be checked for outliers, multicollinearity, and general data quality problems, because these can affect how reliable the results are (Nau, 2024). Since the data is based on a US insurance dataset, it was treated as a proof of concept rather than a final pricing solution for a South African medical aid scheme.

# Plan of analysis

The analysis started with exploratory data analysis to understand the dataset before modelling. This included checking the structure of the data, the variable types, missing values, and duplicate rows. Summary statistics and visuals such as histograms, boxplots, scatterplots, and a pair plot were then used to explore distributions, spot unusual values, and examine how the variables related to medical charges. Müller and Guido (2016) explain that exploring the data before modelling helps reveal patterns and shows whether the available features are likely to support prediction.

The next step was feature preparation and selection. The variables from the brief were used as the starting point, and the categorical variables were encoded so that they could be used in the model. The relationships between predictors were also reviewed to identify which variables seemed more useful and whether any overlapped too much.

After that, the data was split into training and testing sets, and a baseline Linear Regression model was trained. Model performance was then evaluated using MAE, MSE, RMSE, and $R^2$, together with residual analysis, because patterns in the errors can affect how reliable the results are (Nau, 2024). If the first model showed clear weaknesses, retraining would be considered to test whether performance could be improved.

Finally, the report brings the full analysis together by summarising the main steps, findings, and results concisely.

# Exploratory Data Analysis (EDA)

The dataset was explored before model training to check its quality and to understand how the variables relate to medical charges. The data loaded correctly and, after inspection, it was found that the dataset originally contained 1,338 rows and 7 columns. The structure check also showed that the dataset included 4 numeric variables and 3 categorical variables, with no missing values in any column. There were no missing values, but one duplicate row was found and removed, which left 1,337 records for the analysis.

| Check | Result |
|---|---|
| Original shape | 1,338 rows, 7 columns |
| Variable types | 4 numeric, 3 categorical |
| Missing values | 0 |
| Duplicate rows found | 1 |

| | |
|---|---|
| Final shape after cleaning | 1,337 rows, 7 columns |

*Table 1 - Data quality and structure summary*

A condensed summary statistics table was created for the main numeric variables to make the results easier to read and report. It showed that charges, which is the target variable, had a very wide spread, ranging from 1,121.87 to 63,770.43. The mean charge was also noticeably higher than the median, which suggested that the values were not evenly distributed. This pattern was supported by the histogram and boxplot for charges, which showed a clear right-skew and several high-value outliers. In other words, most observations were clustered at lower charge values, while a smaller number of much higher charges pulled the distribution upward.

| Variable | Mean | Median | Minimum | Maximum |
|---|---|---|---|---|
| **Age** | 39,22 | 39 | 18 | 64 |
| **BMI** | 30,66 | 30,4 | 15,96 | 53,13 |
| **Children** | 1,1 | 1 | 0 | 5 |
| **Charges** | 13279,12 | 9386,16 | 1121,87 | 63770,43 |

*Table 2 - Condensed summary statistics for the main numeric variables*

After looking at the summary statistics, the next step was to explore the data visually. This made it easier to see how the numeric variables were distributed and whether any unusual patterns stood out. Visuals were very useful for understanding charges, because the target variable showed a wide spread in the earlier table. They also helped show which variables appeared more closely linked to medical charges, which is important before moving on to model training.

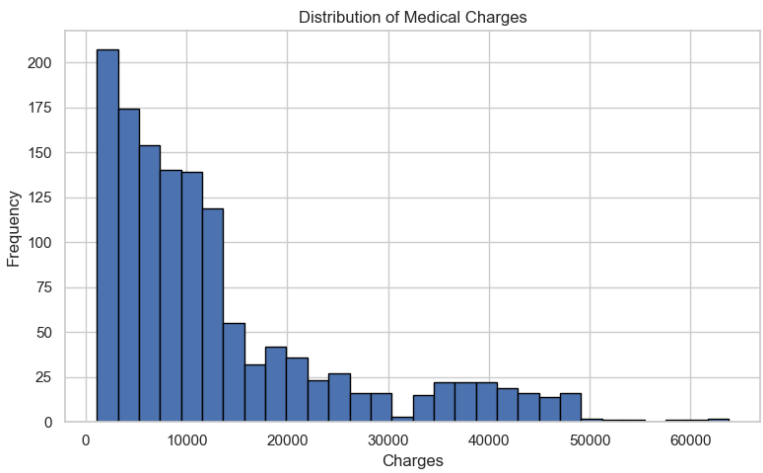



*Figure 1 - Distribution of medical charges*

The histogram of charges showed that the target variable was not evenly distributed. Most of the values were concentrated at the lower end, while a smaller number of much higher charges stretched the distribution to the right. This means the variable is right-skewed, which supports the earlier summary statistics where the mean was higher than the median.

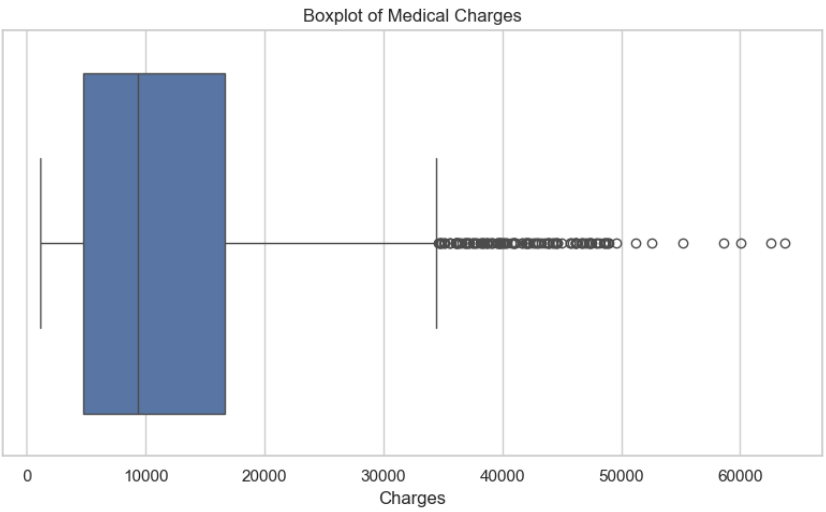



*Figure 2 - Boxplot of medical charges*

The boxplot for charges supported the same finding. It showed several high-value outliers above the main body of the data, which confirms that some individuals had much higher medical charges than the rest of the dataset. This is important because extreme values in the target variable may affect the regression model later.

The visual analysis also helped show which variables seemed more closely related to charges. Age showed a clear positive relationship, which means that charges generally increased as age increased. BMI also appeared to have a positive relationship with charges, although the pattern was weaker and the points were more spread out. The strongest difference was seen in smoking status, where smokers had much higher charges than non-smokers. In comparison, sex, region, and children showed weaker patterns, with a lot of overlap between the groups. This suggests that they may still have some effect, but probably not as much as smoking status, age, or BMI.

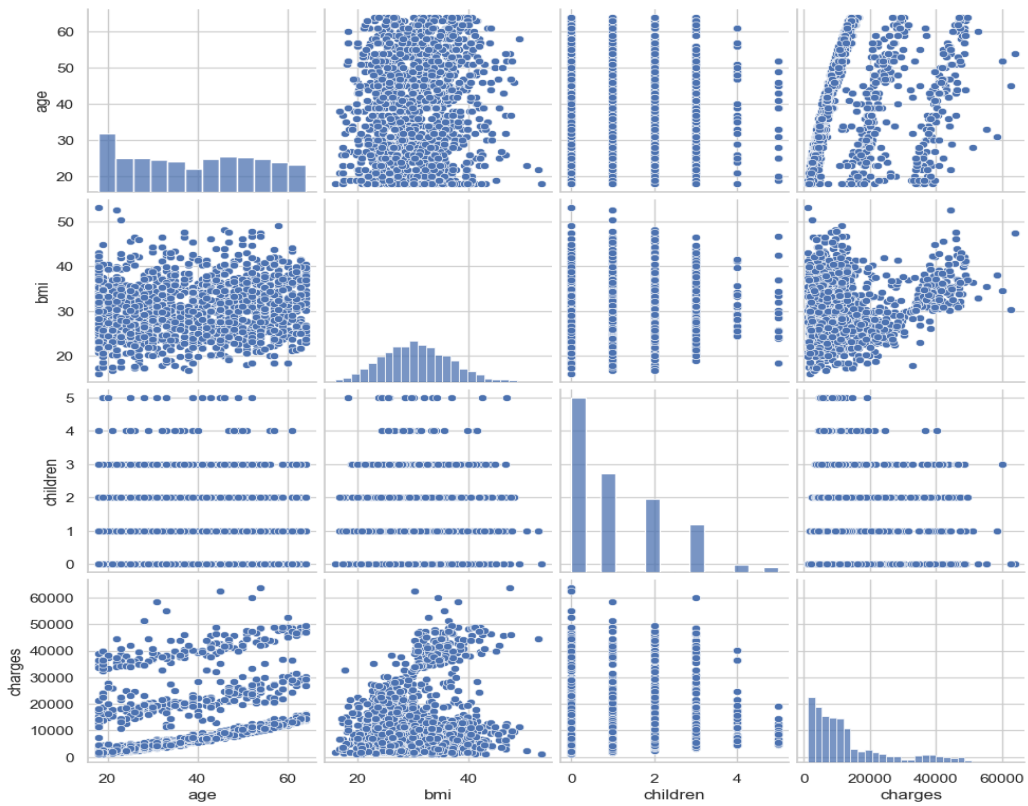



*Figure 3 - Pair plot of the main numeric variables*

The pair plot gives a combined view of the relationships between the main numeric variables and charges. It shows that age has the clearest positive relationship with charges, while BMI also shows a positive but weaker pattern. Children appear to have a much weaker relationship with charges.

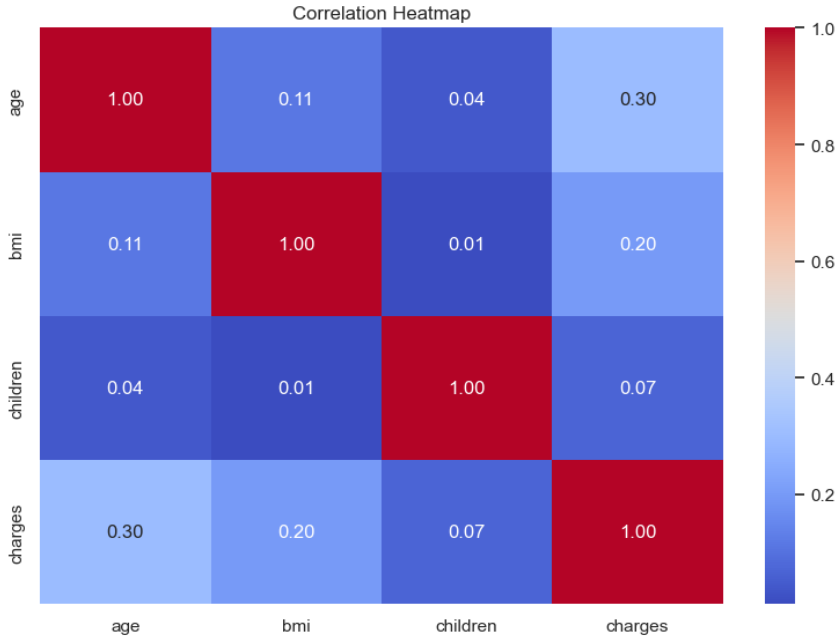



*Figure 4 - Correlation heatmap*

The correlation heatmap supported the visual findings from the earlier plots. Among the numeric variables, age had the strongest relationship with charges, followed by BMI, while children showed only a weak relationship. The heatmap also showed weak relationships between the numeric predictors themselves, which suggests that there was no serious multicollinearity problem at this stage.

The boxplot for smoking status shows the clearest difference in the dataset. Smokers have much higher charges overall than non-smokers, with a higher median and a wider spread. This suggests that smoking status is likely to be one of the most important variables in the Linear Regression model.

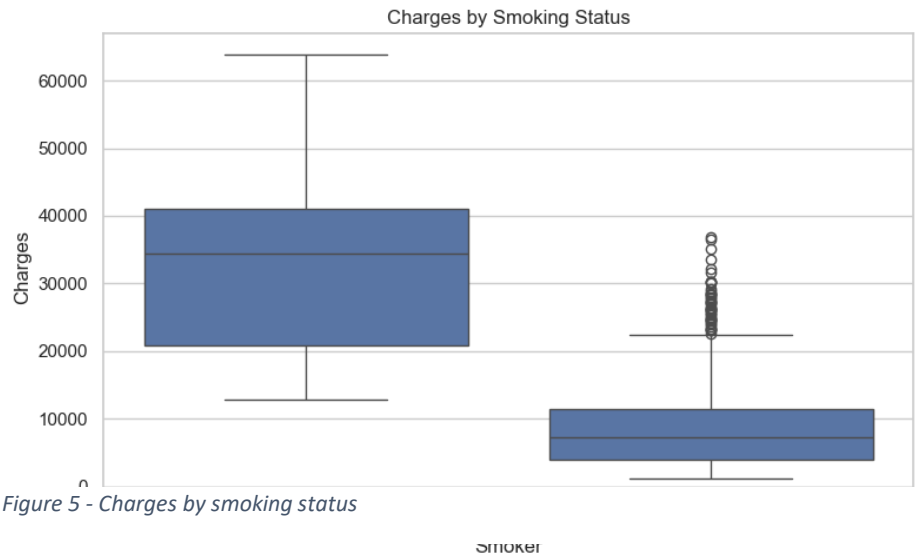



*Figure 5 - Charges by smoking status*

Overall, the EDA showed that the dataset was suitable for modelling and helped identify the variables that seemed most useful for prediction. Smoking status stood out the most, while age and BMI also showed meaningful relationships with charges. The data was mostly clean,

and the main issue found was the wide spread and right-skew in the target variable. These findings helped guide the next step, which was feature preparation and model training.

# Feature selection

The feature selection began with the variables given in the brief, namely age, sex, BMI, number of children, smoking status, and region. These variables were used as the starting point because they are directly related to the problem of predicting medical charges.

Before the model could be trained, the categorical variables sex, smoker, and region had to be converted into numeric form. This was done so that all the input features would be in a format that the Linear Regression model could use. After encoding was done, the dataset contained only numeric input variables, which made it suitable for modelling.

| Feature type | Variables |
|---|---|
| Original numeric features | Age, BMI, Children |
| Encoded categorical features | Sex_male, Smoker_yes, Region_northwest, Region_southeast, Region_southwest |
| Target variable | Charges |

An additional feature check was then carried out using the encoded feature set. This showed that smoking status had the strongest relationship with charges by far, while age and BMI also showed useful positive relationships. The remaining variables had weaker individual relationships with charges, but they were kept in the first model because they were relevant to the business problem and could still contribute when combined with other predictors.

| Feature | Correlation with charges |
|---|---|
| smoker_yes | 0,787 |
| age | 0,298 |
| bmi | 0,198 |
| region_southeast | 0,074 |
| children | 0,067 |
| sex_male | 0,058 |
| region_northwest | -0,039 |
| region_southwest | -0,044 |

*Table 3 - Feature correlation table*

The feature selection process confirmed that the dataset contained a suitable set of predictors for the first Linear Regression model. The EDA and encoded feature check both suggested that smoker, age, and BMI were likely to be the most useful variables, while the other features were retained so that their contribution could be assessed during model training and evaluation.

# Train model

After the features had been prepared, the dataset was split into training and testing sets. The training set was used to fit the model, while the testing set was kept aside so that the model could later be checked on unseen data. An 80/20 split was used, which means most of the data was used for learning, while a smaller portion was reserved for evaluation. A fixed random state “random_state=42” was also used so that the split would stay the same each time the analysis was run.

| Dataset portion | Number of rows |
|---|---|
| Training set | 1069 |
| Testing set | 268 |

*Table 4 - train-test split*

A Linear Regression model was then created and trained on the training data. At this stage, the model learned the relationship between the input features and the target variable, charges, based only on the training set. This allowed the model to build a baseline pattern from the available data before being tested on new cases

After training, the model was used to generate predicted charge values for the test set. A comparison between the actual and predicted values showed that the model was often reasonably close, although there were still some clear differences in certain cases. This was expected, since regression models do not usually predict every value exactly. Even so, the comparison suggested that the model was capturing the general pattern in the data and was ready for formal evaluation.

| Actual Charges | Predicted Charges |
|---|---|
| 8688,86 | 8143,69 |
| 5708,87 | 5737,12 |
| 11436,74 | 14369,31 |
| 38746,36 | 31745,51 |
| 4463,21 | 8962,39 |
| 9304,7 | 13149,72 |
| 38511,63 | 30446,76 |
| 2150,47 | 1453,29 |
| 7345,73 | 10633,02 |
| 10264,44 | 11318,94 |

*Table 5 -Actual and predicted charges for the first 10 test cases*

# Interpret and evaluate model

The model was evaluated using four standard regression measures, namely MAE, MSE, RMSE, and $R^2$. These were used to evaluate how close the predicted charges were to the actual charges and how well the model explained the overall pattern in the data.

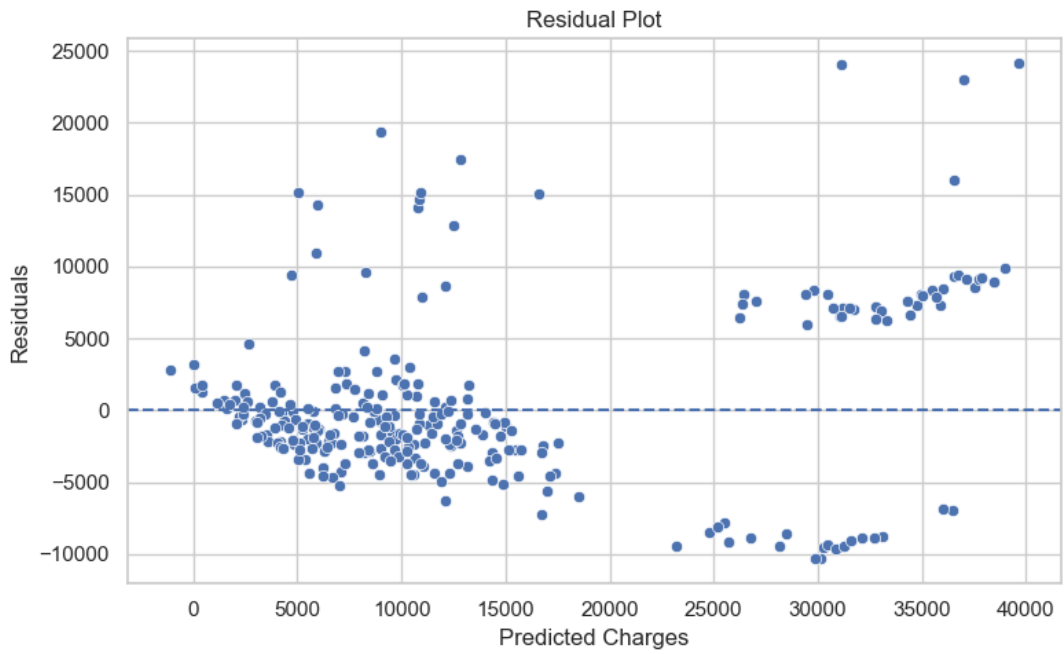
| Metric | Value |
|---|---|
| Mean Absolute Error (MAE) | 4177,05 |
| Mean Squared Error (MSE) | 35478020,68 |
| Root Mean Squared Error (RMSE) | 5956,34 |
| $R^2$ Score | 0,81 |

*Table 6 - Model evaluation results*

The results showed that the model performed well on the test data. The MAE was 4177.05, which means that the model's predictions were off by about R4,177 on average. The RMSE was 5956.34, which was higher than the MAE. This suggests that many predictions were reasonably close, but some errors were noticeably larger. The $R^2$ score was 0.81, which means the model explained about 81% of the variation in medical charges. Overall, this is a good result and shows that the model was able to capture a large part of the relationship between the predictors and charges.

The residual plot gave further insight into the model's performance. The errors were not spread out completely randomly around zero, and the spread became wider at higher predicted

charge values. There were also some large positive and negative residuals. This suggests that the model was better at predicting some cases than others, especially when charges were high. Even though the model performed reasonably well overall, the residual pattern




suggested that the fit was not equally strong across all charge levels, especially for higher values.

*Figure 6 - Residual plot for the Linear Regression model*

# Retraining decision

To test whether the model could be improved, Ridge Regression was used as a retraining step. Ridge Regression was used because it is still a linear model, but it adds L2 regularisation to keep coefficient values under control and reduce overfitting (Rambotti, 2023). Several alpha values were then tested to see whether regularisation would improve the model's performance.

| Alpha | MAE | MSE | RMSE | $R^2$ Score |
|---|---|---|---|---|
| 0,1 | 4178,75 | 35495705,7 | 5957,83 | 0,81 |
| 1 | 4194,01 | 35656881 | 5971,34 | 0,81 |
| 10 | 4344,07 | 37440354,28 | 6118,85 | 0,8 |
| 100 | 5481,37 | 59465107,59 | 7711,36 | 0,68 |

*Table 7 - Ridge alpha results table*

The results showed that the best Ridge model was achieved at alpha = 0.1. However, when this model was compared to the original Linear Regression model, it did not perform better. The Ridge model had slightly higher error values, while the $R^2$ score remained at 0.81. This showed that the original model was still the better option overall.

| Model | MAE | MSE | RMSE | $R^2$ Score |
|---|---|---|---|---|
| Linear Regression | 4177,05 | 35478020,68 | 5956,34 | 0,81 |
| Ridge Regression | 4178,75 | 35495705,7 | 5957,83 | 0,81 |

*Table 8 - Comparison of Linear Regression and Ridge Regression*

For that reason, the original Linear Regression model was kept as the final model for this analysis.

# Concluding remarks

This analysis set out to determine whether a Linear Regression model could be used to predict medical charges from variables such as age, sex, BMI, number of children, smoking status, and region. The results showed that the dataset was suitable for this purpose after basic cleaning and preparation. The EDA also showed that charges was strongly right skewed and that smoking status, age, and BMI were the variables most clearly linked to medical charges.

After encoding the categorical variables and training the model, the baseline Linear Regression model performed well. It explained about 81% of the variation in charges and produced prediction errors that were reasonable for a first model. Although Ridge Regression was tested as a retraining step, it did not improve the results, so the original Linear Regression model was kept as the final model.

It is thus my assertion that a Linear Regression model can be used as a proof-of-concept approach to predict medical charges in this dataset. While the model was not perfect, it captured a large part of the pattern in the data and provided a solid starting point for this type of pricing analysis, while also showing where future improvement may still be possible.

# Reference list

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Rambotti, G. 2023. What is ridge regression? IBM. Available at: https://www.ibm.com/think/topics/ridge-regression [Accessed 31 March 2026].